WORKED SOLUTION

STEP III 1991, Question 7

Pure • Integration Using Symmetry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q7

The key idea

The integral of $\ln(\sin x)$ can't be found from an antiderivative, but symmetry does all the work. The substitution $x \mapsto \frac{1}{2}\pi - x$ swaps sin and cos, and $x \mapsto \pi - x$ shows that $\sin x$ on $[0, \pi]$ is two copies of its values on $[0, \frac{1}{2}\pi]$. Writing $\sin x \cos x = \frac{1}{2}\sin 2x$ then gives an equation that the unknown integral must satisfy. Part (iii) is an integration by parts that turns $x \cot x$ into $\ln(\sin x)$, and part (ii) deals with the boundary term.

Solution

Throughout, write

$$I = \int_0^{\frac{1}{2}\pi} \ln(\sin x)\,\mathrm{d}x.$$

Part (i)

The sine and cosine integrals agree. Substitute $x = \frac{1}{2}\pi - t$, so $\mathrm{d}x = -\mathrm{d}t$ and $\sin x = \cos t$. The limits $0, \frac{1}{2}\pi$ become $\frac{1}{2}\pi, 0$, and the minus sign reverses them back:

$$I = \int_{\frac{1}{2}\pi}^{0} \ln(\cos t)\,(-\mathrm{d}t) = \int_0^{\frac{1}{2}\pi} \ln(\cos t)\,\mathrm{d}t.$$

Relating to $\sin 2x$**.** Adding the two equal integrals and using $\ln a + \ln b = \ln(ab)$ (valid since $\sin x, \cos x > 0$ on the open interval):

$$\begin{aligned} 2I &= \int_0^{\frac{1}{2}\pi} \ln(\sin x \cos x)\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \ln(\tfrac{1}{2}\sin 2x)\,\mathrm{d}x \\ &= \int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x - \tfrac{1}{2}\pi \ln 2. \end{aligned}$$

Halving gives $I = \frac{1}{2}\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x - \frac{1}{4}\pi \ln 2$, which completes the first chain of equalities.

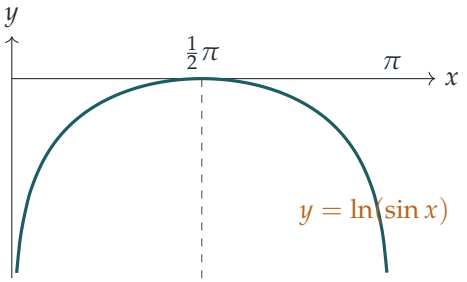
The second chain. With $u = 2x$, $\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$, and the limits become 0 and π:

$$\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x = \tfrac{1}{2}\int_0^{\pi} \ln(\sin u)\,\mathrm{d}u.$$

Now split $[0, \pi]$ at $\frac{1}{2}\pi$. On the second half substitute $u = \pi - v$. Since $\sin(\pi - v) = \sin v$ and the limits $\frac{1}{2}\pi, \pi$ become $\frac{1}{2}\pi, 0$,

$$\int_{\frac{1}{2}\pi}^{\pi} \ln(\sin u)\,\mathrm{d}u = \int_0^{\frac{1}{2}\pi} \ln(\sin v)\,\mathrm{d}v = I.$$

The graph below shows this symmetry: the curve $y = \ln(\sin x)$ is symmetric about $x = \frac{1}{2}\pi$.



So $\int_0^{\pi} \ln(\sin u)\,\mathrm{d}u = 2I$, and therefore

$$\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x = \tfrac{1}{2}\int_0^{\pi} \ln(\sin x)\,\mathrm{d}x = \tfrac{1}{2}(2I) = I,$$

as required.

Evaluating I. Putting the second result into the first gives

$$I = \tfrac{1}{2}I - \tfrac{1}{4}\pi\ln 2 \quad\Longrightarrow\quad I = \int_0^{\frac{1}{2}\pi} \ln(\sin x)\,\mathrm{d}x = -\tfrac{1}{2}\pi\ln 2.$$

This step needs I to be a finite number, which is why the question lets us assume convergence. Without that, "$I = \frac{1}{2}I + c$" would prove nothing.

Part (ii)

Let $x \geq 1$ and put $u = \sqrt{x}$. Then $u \geq 1$, so the given inequality applies:

$$\ln\sqrt{x} < \sqrt{x}, \qquad \text{that is,} \qquad \tfrac{1}{2}\ln x < \sqrt{x}.$$

First limit. For $x \geq 1$ we have $\ln x \geq 0$, and dividing $\ln x < 2\sqrt{x}$ by $x > 0$ gives

$$0 \leq \frac{\ln x}{x} < \frac{2}{\sqrt{x}}.$$

The right-hand side tends to 0 as $x \to \infty$, so by the squeeze (sandwich) principle $\dfrac{\ln x}{x} \to 0$.

Second limit. For $0 < x < 1$ put $x = 1/t$, so $t > 1$ and $t \to \infty$ as $x \to 0^+$. Then

$$x \ln x = \frac{1}{t} \ln \frac{1}{t} = -\frac{\ln t}{t} \to 0$$

by the first limit. So $x \ln x \to 0$ as $x \to 0$ through positive values.

Part (iii)

Note first that the integrand is harmless at 0: $x \cot x = \dfrac{x}{\sin x} \cos x \to 1$. Since $\cot x$ is the derivative of $\ln(\sin x)$ on $(0, \frac{1}{2}\pi)$, we integrate by parts with $u = x$ and $\mathrm{d}v = \cot x \, \mathrm{d}x$, $v = \ln(\sin x)$. The logarithm blows up at 0, so we work on $[\varepsilon, \frac{1}{2}\pi]$ and then let $\varepsilon \to 0^+$:

$$\int_\varepsilon^{\frac{1}{2}\pi} x \cot x \, \mathrm{d}x = \Big[x \ln(\sin x) \Big]_\varepsilon^{\frac{1}{2}\pi} - \int_\varepsilon^{\frac{1}{2}\pi} \ln(\sin x) \, \mathrm{d}x.$$

At the top limit $\ln(\sin \frac{1}{2}\pi) = \ln 1 = 0$. So the boundary term is just $-\varepsilon \ln(\sin \varepsilon)$.

> **Ask yourself** Why does $\varepsilon \ln(\sin \varepsilon)$ tend to 0? Part (ii) is about $x \ln x$, not $x \ln(\sin x)$.

Write $s = \sin \varepsilon$, which tends to 0 through positive values as $\varepsilon \to 0^+$. Then

$$\varepsilon \ln(\sin \varepsilon) = \frac{\varepsilon}{\sin \varepsilon} \cdot s \ln s.$$

The first factor tends to 1 (the standard limit $\sin \varepsilon / \varepsilon \to 1$), and $s \ln s \to 0$ by part (ii). So the boundary term tends to 0.

Letting $\varepsilon \to 0^+$ and using part (i):

$$\int_0^{\frac{1}{2}\pi} x \cot x \, \mathrm{d}x = 0 - I = \tfrac{1}{2}\pi \ln 2.$$

Checks

- Sign: $\ln(\sin x) \le 0$ on the interval, so I must be negative, and $-\frac{1}{2}\pi \ln 2 \approx -1.0888$ is. On the other hand $x \cot x > 0$ on $(0, \frac{1}{2}\pi)$, and our answer is positive.
- Size: $x \cot x$ falls from 1 to 0 across an interval of length $\frac{1}{2}\pi \approx 1.571$, so an integral of about 1.09 is plausible.
- Numerically, midpoint quadrature gives $\int_0^{\frac{1}{2}\pi} x \cot x \, \mathrm{d}x \approx 1.0887930$, matching $\frac{1}{2}\pi \ln 2 = 1.0887930\ldots$, and $\int_0^{\frac{1}{2}\pi} \ln(\sin x) \, \mathrm{d}x \approx -1.08879$.

Common slips

- Forgetting the factor $\frac{1}{2}$ from $\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$ when changing to $u = 2x$, which gives $\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x = 2I$ and leads nowhere.
- In part (iii), writing $[x\ln(\sin x)]_0^{\frac{1}{2}\pi} = 0$ without justifying the limit at 0. That limit is exactly what part (ii) is for.

Beyond STEP

Both integrals here are *improper*: their integrands, or the antiderivatives used, are unbounded at an endpoint. They are defined as limits, as we did with ε. A first analysis course makes this precise and gives tests for when such limits exist. These are the tests behind the question's "you may assume" clause. The same symmetry tricks evaluate many more integrals, such as $\int_0^{\pi/2} \ln(\tan x)\,\mathrm{d}x = 0$. Abbott's *Understanding Analysis* covers the theory of limits and integration that underpins all of this.

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